

MAT1002 Lecture 6,

Outline

- Binomial Series
- Taylor's theorem (Big and little O) and applications
- Euler's formula
- Proof of Taylor's theorem

Binomial Series

Consider $f(x) := (1+x)^\alpha$, where $\alpha \in \mathbb{R}$. Then $x=0$

$$f'(x) = \alpha(1+x)^{\alpha-1} // \Rightarrow f'(0) = \alpha //$$

$$f''(x) = \alpha(\alpha-1)(1+x)^{\alpha-2} // \Rightarrow f''(0) = \alpha(\alpha-1) //$$

$\vdots$

$$f^{(n)}(x) = \alpha(\alpha-1)\dots(\alpha-n+1)(1+x)^{\alpha-n} // \Rightarrow f^{(n)}(0) = \alpha(\alpha-1)\dots(\alpha-n+1) //$$

Maclaurin series of $f(x) = (1+x)^\alpha$ is $\sum_{n=0}^{\infty} \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!} x^n$

Def:

• For any $\alpha \in \mathbb{R}$ and $n \in \mathbb{N} := \{0, 1, 2, \dots\}$,

"α choose n" $\rightarrow$ $\binom{\alpha}{n} := \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!}$

is a binomial coefficient. By default, $\binom{\alpha}{0} = 1$.

• The series $\sum_{n=0}^{\infty} \binom{\alpha}{n} x^n$ is a binomial series.

Remarks

When $\alpha \in \mathbb{N}$, $\binom{\alpha}{n}$ is the "usual" binomial coefficient that has a counting meaning. In particular, $\binom{\alpha}{n} = 0$ for all $n > \alpha$.

$$(1+x)^m = \dots + x^m \\ = 1 + a_0 x + a_1 x^2 + \dots + a_{m-1} x^{m-1}$$

Convergence

If $\alpha = m \in \mathbb{N}$, then the binomial series is a finite sum

$$\sum_{n=0}^m \binom{m}{n} x^n,$$

$$(1+x)^{\frac{1}{2}} = \sqrt{1+x}$$

hence converging for all $x \in \mathbb{R}$.

If $\alpha \notin \mathbb{N}$, then the binomial series is infinite. Since

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{\alpha(\alpha-1)\dots(\alpha-n)}{(n+1)!} \cdot \frac{n!}{\alpha(\alpha-1)\dots(\alpha-n+1)} \right| |x| = \frac{|\alpha-n|}{n+1} |x| \\ = |x| \frac{\left| \frac{\alpha}{n} - 1 \right|}{\left| 1 + \frac{1}{n} \right|} \rightarrow \underline{|x|} \text{ as } n \rightarrow \infty.$$

By Ratio Test, radius of convergence is $R=1$.

Fact: $\sum_{n=0}^{\infty} \binom{\alpha}{n} x^n = (1+x)^{\alpha} \text{ for } x \in (-1, 1).$

Proof: Ex. 58 in 10.10.

e.g. For $\alpha = \frac{1}{2}$, $\binom{\frac{1}{2}}{0} = 1$, $\binom{\frac{1}{2}}{1} = \frac{1}{2}$, $\binom{\frac{1}{2}}{2} = \frac{\frac{1}{2}(\frac{1}{2}-1)}{2!} = -\frac{1}{8}$,

and for $n \geq 2$,

$$\binom{\frac{1}{2}}{n} = \frac{\frac{1}{2}(\frac{1}{2}-1)\dots(\frac{1}{2}-n+1)}{n!} = \frac{\frac{1}{2}(-\frac{1}{2})\dots(-\frac{2n-3}{2})}{n!} \\ = (-1)^{n-1} \frac{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-3)}{2^n \cdot n!}$$

These give the coefficients for the binomial series for

$$f(x) = \sqrt{1+x} = (1+x)^{\frac{1}{2}}.$$

Frequently Used Maclaurin Series that Converge to Their Functions

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \cdots = \sum_{n=0}^{\infty} x^n, \quad |x| < 1$$

$$\frac{1}{1+x} = 1 - x + x^2 - \cdots + (-x)^n + \cdots = \sum_{n=0}^{\infty} (-1)^n x^n, \quad |x| < 1$$

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad |x| < \infty$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad |x| < \infty$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad |x| < \infty$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^{n-1} \frac{x^n}{n} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}, \quad -1 < x \leq 1$$

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^n \frac{x^{2n+1}}{2n+1} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}, \quad |x| \leq 1$$

We only discuss
 $|x| < 1$; see
 10.10 for
 $|x| \leq 1$.

Big Oh and Little Oh (as $x \rightarrow a$)

Def: For functions f and g , we write

• $f(x) = \underset{\text{Little Oh}}{o}(g(x))$ as $x \rightarrow a$, if $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 0$.

• $f(x) = \underset{\text{Big Oh}}{O}(g(x))$ as $x \rightarrow a$, if $\exists M \in \mathbb{R}$ and $\exists \delta > 0$ s.t.

$$\forall x \in (a-\delta, a+\delta) \setminus \{a\}, \quad |f(x)| \leq M|g(x)|.$$

$\frac{1}{\ln|x-a|} \rightarrow 0$

$$\Leftrightarrow \frac{|f(x)|}{|g(x)|} \leq M \quad \text{if } g \text{ is never zero}$$

e.g.

If $m, n \in \mathbb{N}$ and $m > n$, then $x^m = o(x^n)$ as $x \rightarrow 0$, since

$$\lim_{x \rightarrow 0} \frac{x^m}{x^n} = \lim_{x \rightarrow 0} x^{m-n} = 0$$

Can also say $x^m = O(x^n)$, since $\left| \frac{x^m}{x^n} \right| = |x^{m-n}| \leq 1$ for x with $|x| \leq 1$.

Taylor's theorem : Revisit

Suppose that on $I = \underbrace{(a-\delta, a+\delta)}_{\text{a neighbourhood of } a}$, $|f^{(n+1)}|$ is bounded by M .

Then by Taylor's theorem, for any $x \in I$,

$$\begin{aligned} f(x) &= \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + R_n(x) \\ &= f(a) + f'(a)(x-a) + \dots + \frac{f^{(n)}(a)}{n!} (x-a)^n + \overset{\text{big}}{O((x-a)^{n+1})} \text{ as } x \rightarrow a, \end{aligned}$$

$$\text{Since } \left| \frac{R_n(x)}{(x-a)^{n+1}} \right| = \frac{|f^{(n+1)}(c)|}{(n+1)!} \leq \frac{M}{(n+1)!} \text{ for all } x \in I \setminus \{a\}.$$

On the other hand, $R_n(x) = \overset{\text{little}}{o}((x-a)^n)$ as $x \rightarrow a$. Indeed,

$$\left| \frac{R_n(x)}{(x-a)^n} \right| = \frac{|f^{(n+1)}(c)|}{(n+1)!} |x-a| \leq \frac{M}{(n+1)!} |x-a| \rightarrow 0 \text{ as } x \rightarrow a,$$

$$\text{So } \lim_{x \rightarrow a} \frac{R_n(x)}{(x-a)^n} = 0.$$

We summarize the statement above in the following theorem.

Taylor's Theorem (O and o version)

Suppose $f, f', \dots, f^{(n)}$ are continuous on a neighbourhood I of a , and $f^{(n+1)}$ exists and is bounded on I . Then, as $x \rightarrow a$,

$$f(x) = \underbrace{\sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k}_{P_n(x)} + O((x-a)^{n+1}) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + o((x-a)^n).$$

The last line above also holds for all n if f can be represented by its Taylor series on some neighbourhood of a .

Theorem If f can be represented by its Taylor series with center $x=a$ on some neighbourhood I of a , then for any fixed $n \geq 0$,

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + O((x-a)^{n+1}), \text{ as } x \rightarrow a.$$

Proof: $\forall x \in I \setminus \{a\}$,

$$R_n(x) = f(x) - \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k = \sum_{k=n+1}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k,$$

$$\text{So } \frac{R_n(x)}{(x-a)^{n+1}} = \frac{f^{(n+1)}(a)}{(n+1)!} + \frac{f^{(n+2)}(a)}{(n+2)!} (x-a) + \dots$$

$$\Rightarrow \lim_{x \rightarrow a} \frac{R_n(x)}{(x-a)^{n+1}} = \frac{f^{(n+1)}(a)}{(n+1)!}$$

$\Rightarrow \left| \frac{R_n(x)}{(x-a)^{n+1}} \right|$ is bounded for x sufficiently close to a

$$\Rightarrow R_n(x) = O((x-a)^{n+1}) \quad \text{as } x \rightarrow a.$$

□

Remarks about Big-O (as $x \rightarrow 0$)

(a) If $p > 0$, then $\lim_{x \rightarrow 0} O(x^p) = 0$. (Sandwich)

(b) If $f(x) = O(x^m)$ and $g(x) = O(x^n)$, then

$$\begin{aligned} & \left(\begin{array}{l} O(x^m) O(x^n) \\ = O(x^{m+n}) \end{array} \right) \quad f(x)g(x) = O(x^{m+n}) \quad \text{and} \quad f(x)+g(x) = O(x^{\min\{m,n\}}) \end{aligned}$$

$$\begin{aligned} & O(x^m) + O(x^n) \\ & = O(x^{\min\{m,n\}}) \end{aligned}$$

$$\hookrightarrow |f(x)| \leq M|x^m|, \quad |g(x)| \leq N|x^n| \Rightarrow |f(x)g(x)| \leq MN|x^{m+n}|.$$

If $m > n$, then

$$\begin{aligned} |f(x)+g(x)| & \leq M|x^m| + N|x^n| = (M|x|^{m-n} + N)|x|^n \\ & \leq (M+N)|x|^n \quad \text{if } |x| \leq 1. \end{aligned}$$

The remarks and theorem above may be useful in finding limits.

$$\text{e.g. } \lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right) = \lim_{x \rightarrow 0} \frac{x - \sin x}{x \sin x}$$

$$= \lim_{x \rightarrow 0} \frac{x - (x + O(x^3))}{x(x + O(x^3))} \quad \text{as } x \rightarrow 0$$

$$= \lim_{x \rightarrow 0} \frac{O(x^3)}{x^2 + O(x^4)} = \lim_{x \rightarrow 0} \frac{O(x)}{1 + O(x^2)} = \frac{0}{1+0} = 0.$$

e.g. $\lim_{x \rightarrow 0} \frac{\cos x - 1 + \frac{1}{2}x \sin x}{[\ln(1+x)]^4}$

$$= \lim_{x \rightarrow 0} \frac{1 - \frac{x^2}{2!} + \frac{x^4}{4!} + O(x^6) - 1 + \frac{1}{2}x(x - \frac{x^3}{3!} + O(x^5))}{[x + O(x^2)]^4} \quad \text{as } x \rightarrow 0$$

$$= \lim_{x \rightarrow 0} \frac{\frac{x^4}{4!} - \frac{x^4}{3! \cdot 2} + \frac{1}{2}xO(x^5) + O(x^6)}{[x + O(x^2)]^4} \quad O(x^6) = x^4 O(x^2)$$

$$= \lim_{x \rightarrow 0} \frac{(\frac{1}{4!} - \frac{1}{3! \cdot 2}) + O(x^2)}{(1 + \frac{1}{x}O(x^2))^4} \quad \div x^4 \text{ top \& bottom}$$

$$= \lim_{x \rightarrow 0} \frac{(\frac{1}{4!} - \frac{1}{3! \cdot 2}) + O(x^2)}{(1 + O(x))^4} = \frac{1}{4!} - \frac{1}{3! \cdot 2} = \frac{1}{24} - \frac{1}{12}$$

$$= -\frac{1}{24}.$$

Euler's Formula set: $\mathbb{C}$

A **complex number** is a number of the form $x+yi$, where $x, y \in \mathbb{R}$, and i satisfies $i^2 = -1$. Multiplications are done as usual: $(x+yi)(a+bi) = ax + xbi + yai + ybi^2 = (ax - by) + (bx + ay)i$

The **complex exponential function** $\exp: \mathbb{C} \rightarrow \mathbb{C}$ is defined by

Also written as e^z $\exp(z) := 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots = \sum_{n=0}^{\infty} \frac{z^n}{n!}$

Since $i^0=1$, $i^1=i$, $i^2=-1$, $i^3=-i$, $i^4=-i^2=1=i^0$,

for $\theta \in \mathbb{R}$, we have

$$\begin{aligned} \exp(i\theta) &= 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \frac{\theta^4}{4!} + \frac{i\theta^5}{5!} - \frac{\theta^6}{6!} - \frac{i\theta^7}{7!} + \dots \\ &= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \dots\right) \\ &= \cos \theta + i \sin \theta \end{aligned}$$

Euler's formula: $e^{i\theta} = \cos \theta + i \sin \theta$.

The book takes it as a definition.

One consequence of Euler's formula is the following identity:

if $\theta = \pi$, we have $e^{i\pi} = -1 + 0i$, or

$$e^{i\pi} + 1 = 0,$$

connecting the five constants e , i , π , 1 and 0 .

Proof of Taylor's Theorem

Taylor's Theorem

Suppose:

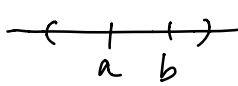
- $f, f', \dots, f^{(n)}$ are continuous on $[a, b]$ (or $[b, a]$);
- $f^{(n+1)}$ exists on (a, b) (or (b, a)).

Then there exists $c \in (a, b)$ (or $c \in (b, a)$) such that

$$f(b) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (b-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!} (b-a)^{n+1}.$$

$$P_n'(x)$$

Proof: • We show for $b > a$. It is similar for $b < a$.

• Define $P_n(x) := \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k$ on $[a, b]$. 

Note that $P_n(a) = f(a)$, $P_n'(a) = f'(a)$, $P_n''(a) = f''(a)$, ...,

$P_n^{(k)}(a) = f^{(k)}(a)$ for all $k \in \{0, \dots, n\}$.

A property of Taylor's polynomial

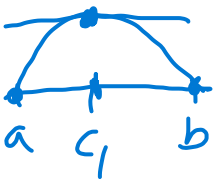
$$\phi_n(a) = f(a) \quad \phi_n(b) = f(b)$$

• Define $\phi_n(x) := P_n(x) + K(x-a)^{n+1}$, where K is chosen such that $f(b) = \phi_n(b)$.

• Note that for $k \in \{0, 1, \dots, n\}$, $\phi_n^{(k)}(a) = P_n^{(k)}(a) = f^{(k)}(a)$.

• Define $F(x) := f(x) - \phi_n(x)$. Then $F^{(k)}(a) = 0$, $\forall k \in \{0, \dots, n\}$,

$$\text{and } F(b) = 0. \quad \begin{aligned} F(a) &= f(a) - \phi_n(a) = f(a) - f(a) = 0 \\ F(b) &= f(b) - \phi_n(b) = 0 \end{aligned}$$



• Since $F(a) = F(b) = 0$, F is continuous on $[a, b]$, diff'able on (a, b) , by Rolle's theorem, $\exists c_1 \in (a, b)$ s.t. $F'(c_1) = 0$.

• Since $F'(a) = F'(c_1) = 0$, F' is continuous on $[a, c_1]$ and diff'able on (a, c_1) , $\exists c_2 \in (a, c_1)$ s.t. $F''(c_2) = 0$.

• Continued with the same logic, $\exists c_n \in (a, c_{n-1})$ s.t. $F^{(n)}(c_n) = 0$.

• Finally, $F^{(n)}$ is continuous on $[a, c_n]$ and diff'able on (a, c_n) ,

so $\exists c \in (a, c_n) \subseteq (a, b)$ s.t. $F^{(n+1)}(c) = 0$.

• But $F^{(n+1)}(x) = f^{(n+1)}(x) - \phi_n^{(n+1)}(x)$

$\times^{n+1}$

0
||

$$\begin{aligned} &= \underline{f^{(n+1)}(x)} - \underline{P_n^{(n+1)}(x)} - \underline{(K(x-a)^{n+1})^{(n+1)}} \\ &= \underline{f^{(n+1)}(x)} - \underline{K(n+1)!}, \end{aligned}$$

So

$$0 = \underline{F^{(n+1)}(c)} = \underline{f^{(n+1)}(c) - K(n+1)!},$$

whence

$$K = \frac{f^{(n+1)}(c)}{(n+1)!}.$$

